Experiment Number: 3 MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

PROGRAM:

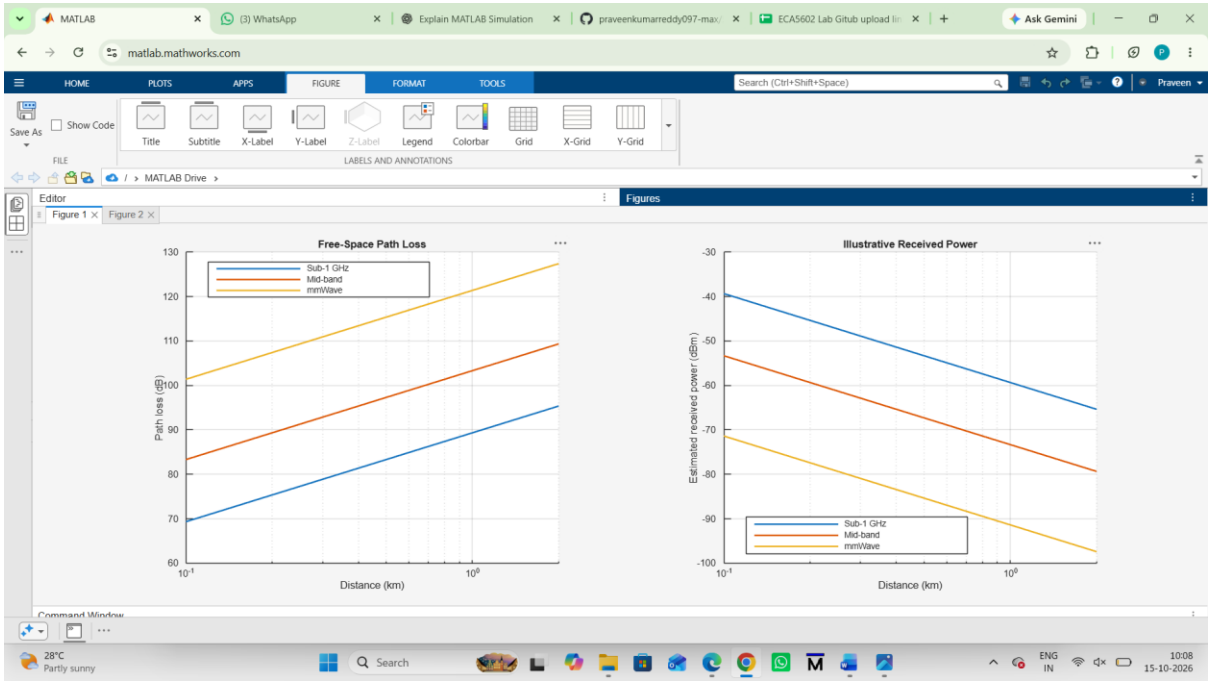
```
% Experiment 3: Illustrative Propagation and Frequency Reuse

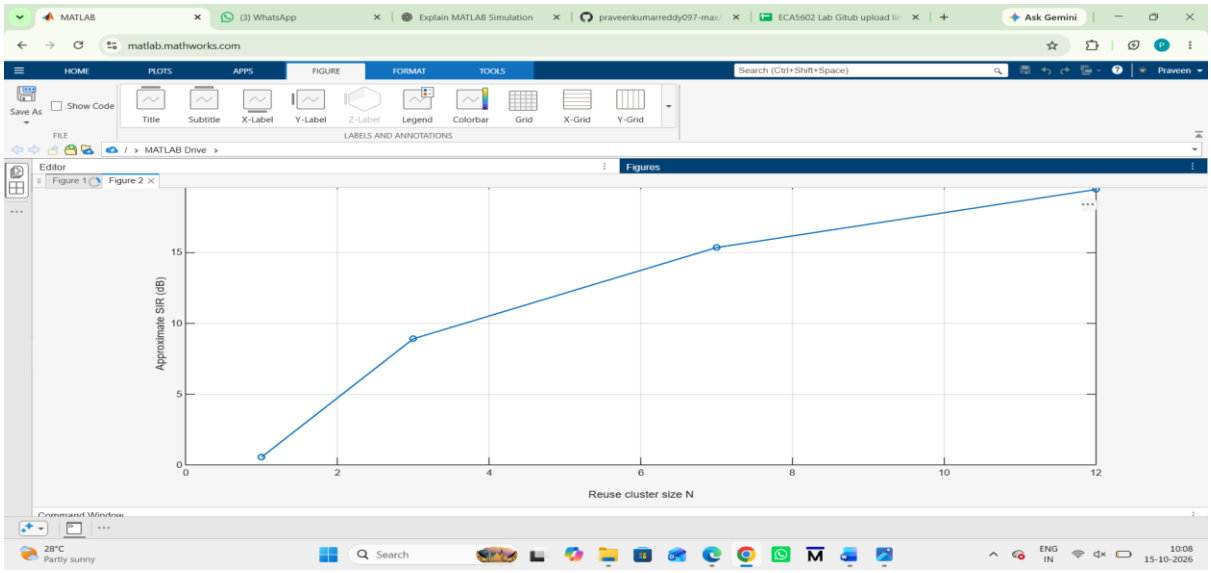
clear;
clc;
close all;
%% 1. Representative frequency bands
freqGHz = [0.7; 3.5; 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0;
grDbi = 0;
distanceKm = logspace(log10(0.1), log10(2), 200);
distanceAtPointKm = 0.5;
%% 2. Free-space path loss model
% d in km, f in MHz
fspl = @(fMHz, dKm) ...
    32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz, distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table( ...
    bandName, freqGHz, lossAtPoint, rxAtPoint, ...
    'VariableNames', ...
    {'Band', 'Frequency_GHz', 'FSPL_dB', 'EstimatedRx_dBm'});
disp('LINK BUDGET AT 0.5 km');
disp(linkTable);
%% 3. Plot free-space path loss and received power
figure('Color', 'w', 'Name', 'Propagation Analysis');
tiledlayout(1, 2, ...
    'TileSpacing', 'compact', ...
    'Padding', 'compact');
% Free-space path loss
nexttile;
hold on;
grid on;
for k = 1:numel(freqMHz)
    plot(distanceKm, fspl(freqMHz(k), distanceKm), ...
        'LineWidth', 1.5);
end
```

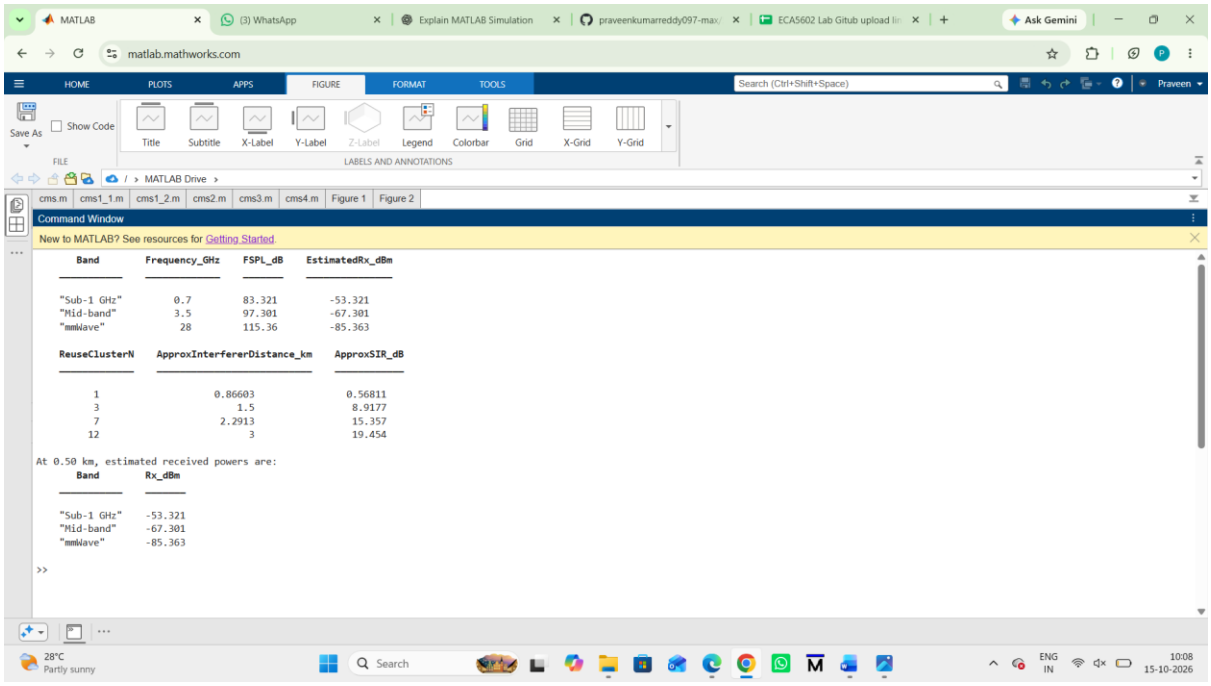
```
set(gca, 'XScale', 'log');
xlabel('Distance (km)');
ylabel('Path loss (dB)');
legend(bandName, 'Location', 'northwest');
title('Free-Space Path Loss');
hold off;
% Estimated received power
nexttile;
hold on;
grid on;
for k = 1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi ...
        - fspl(freqMHz(k), distanceKm);
    plot(distanceKm, rx, 'LineWidth', 1.5);
end
set(gca, 'XScale', 'log');
xlabel('Distance (km)');
ylabel('Estimated received power (dBm)');
legend(bandName, 'Location', 'southwest');
title('Illustrative Received Power');
hold off;
%% 4. Simplified co-channel frequency reuse
cellRadiusKm = 0.5;
reuseN = [1; 3; 7; 12];
pathLossExponent = 3.5;
numberOfInterferers = 6;
% Hexagonal cellular geometry
reuseDistance = sqrt(3 .* reuseN) .* cellRadiusKm;
% Approximate SIR:
% SIR = (D/R)^n / number of equal-distance interferers
sirLinear = (reuseDistance ./ cellRadiusKm) ...
    .^ pathLossExponent / numberOfInterferers;
sirDb = 10*log10(sirLinear);
reuseTable = table( ...
    reuseN, reuseDistance, sirDb, ...
    'VariableNames', ...
    {'ReuseClusterN', 'ReuseDistance_km', 'ApproxSIR_dB'});
disp('FREQUENCY REUSE ANALYSIS');
disp(reuseTable);
%% 5. Plot SIR versus reuse cluster size
figure('Color', 'w');
plot(reuseN, sirDb, 'o-', 'LineWidth', 1.5);
grid on;
```

```
xlabel('Reuse cluster size N');
ylabel('Approximate SIR (dB)');
title('Co-Channel SIR versus Reuse Cluster Size');
%% 6. Display received power summary
fprintf('\nAt %.2f km, estimated received powers are:\n', ...
    distanceAtPointKm);
rxTable = table( ...
    bandName, rxAtPoint, ...
    'VariableNames', {'Band', 'Rx_dBm'});
disp(rxTable);
```

OUTPUT:






MATLAB
(3) WhatsApp
Explain MATLAB Simulation
praveenkumarreddy097-max/
ECA5602 Lab Gitub upload lin
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Praveen
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
/ › MATLAB Drive ›
cms.m
cms1_1.m
cms1_2.m
cms2.m
cms3.m
cms4.m
Figure 1
Figure 2
Command Window
New to MATLAB? See resources for Getting Started.
Band Frequency_GHz FSPL_dB EstimatedRx_dBm
"Sub-1 GHz" 0.7 83.321 -53.321
"Mid-band" 3.5 97.301 -67.301
"mmWave" 28 115.36 -85.363
ReuseClusterN ApproxInterfererDistance_km ApproxSIR_dB
1 0.86603 0.56811
3 1.5 8.9177
7 2.2913 15.357
12 3 19.454
At 0.50 km, estimated received powers are:
Band Rx_dBm
"Sub-1 GHz" -53.321
"Mid-band" -67.301
"mmWave" -85.363
>>
28°C
Partly sunny
Search
ENG
IN
10:08
15-10-2026